Patents: from filing to post-grant challenges

History of patents, from filing to opposing a patent in different jurisdictions

Innovation Management · A.Y. 2026/27
Prof. Grazia Sveva Ascione · Monday 12 October 2026

Politecnico di Torino

Today's lecture

01 A short history of patents

02 Filing a patent

03 What does a patent document look like?

04 Patent families

05 Some figures from the EPO

06 Challenging a patent (after grant)

07 Spoiler of next lecture

01

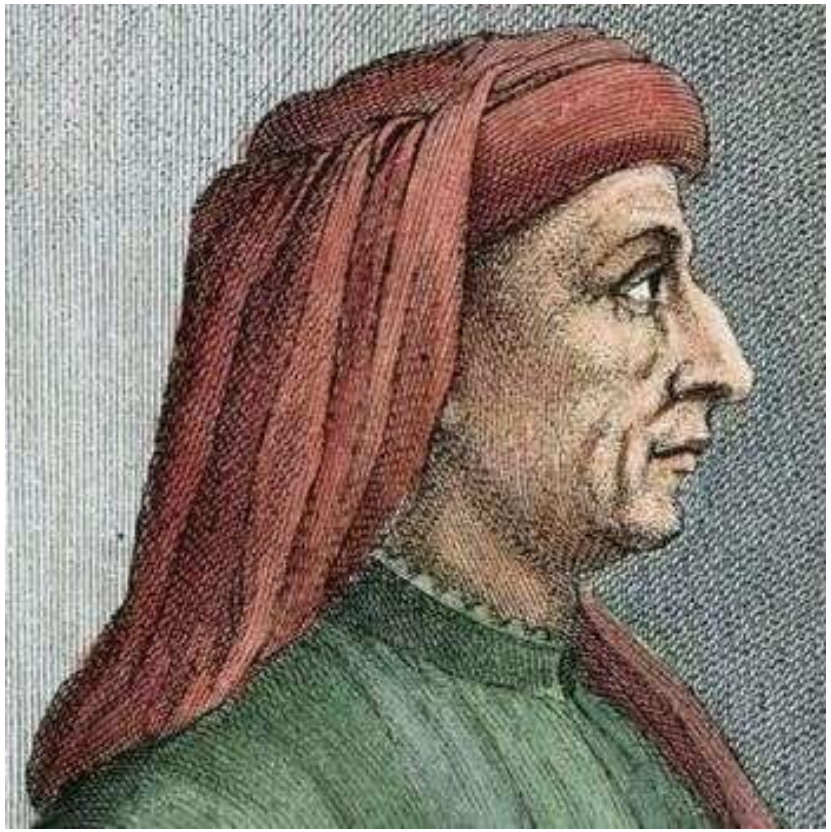
A short history of patents

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The first *privilegium*



The invention (the dome)

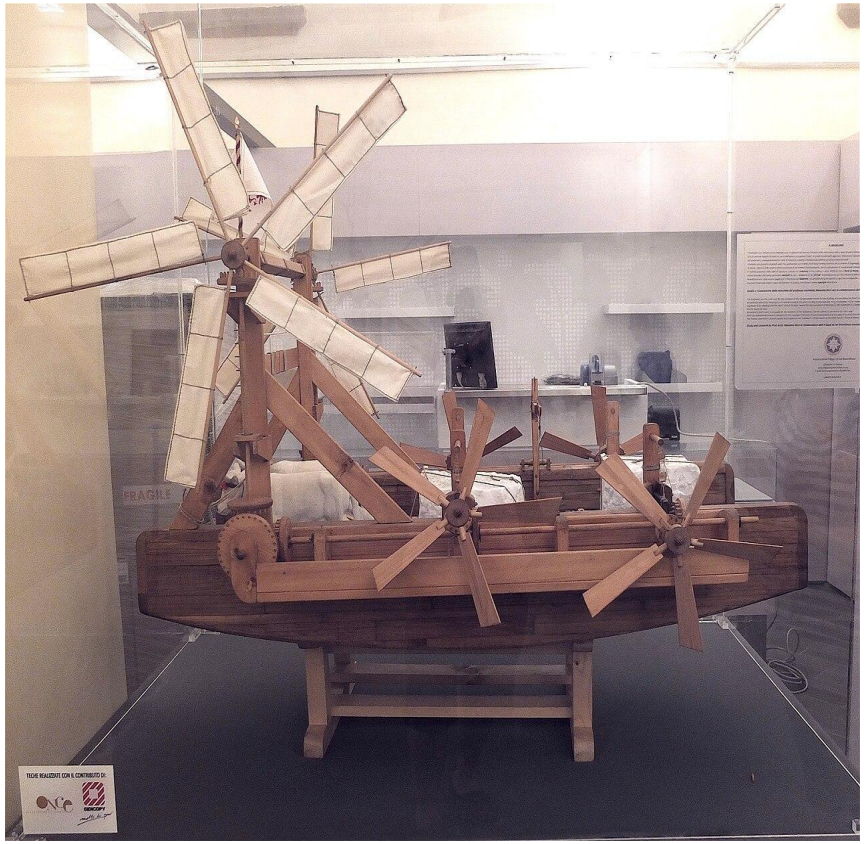


Brunelleschi (inventor)



Ghiberti (potential free rider)

The first patent of invention: Brunelleschi, Florence 1421



The problem	Brunelleschi, architect of the dome of Florence cathedral, had to bring marble from Carrara along the shallow river Arno.
The invention	A special barge for shallow water, nicknamed “Il Badalone”, the monster.
Why a patent	He feared that rivals such as Ghiberti would copy it. For earlier inventions he kept the **secret** by having each component built by a different artisan: impossible for a ship.
The grant (19 June 1421)	In exchange for disclosure, 3 years in which nobody could build or use any new ship on the Arno; ships already known were not affected.
The ending	The first voyage came in 1428, after the privilege had expired. The barge sank near Empoli with 100 tons of marble: Brunelleschi lost about a third of his wealth.

The first patent law: Venice, 1474

Passed by the Venetian Senate on 19 March 1474: 116 votes for, 10 against, 3 abstentions.

For the first time inventors have a claim to a patent: it is no longer a favour of the sovereign.

Preamble	Encourage innovation and attract foreign know-how	→ **The rationale of patents**
Requirements	"***Nuovo et ingegnoso***", and brought "a perfection"	→ **Novelty, inventive step, reduction to practice**
Procedure	Application to the Provveditori, a commercial office	→ **Filing at a patent office**
Rights	10 years of exclusivity	→ **The patent term**
Enforcement	Fine of 100 ducats (about 346 g of gold) and destruction of the device	→ **Damages and corrective orders**
Exception	Use by the State	→ **Public-interest exceptions**

02

Filing a patent

From the inventor’s idea to a filed application

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From the idea to a patent application

1 · The inventor

Has the **idea** and writes an *invention disclosure*: what the invention does, how it works, what is new, what already exists.

- Usually an employee: the disclosure goes to the firm's **IP department** (or a university's **technology transfer office-TTO**)
- The inventor is always named on the patent

2 · The patent attorney

A qualified professional (in Europe: European patent attorney) who drafts the application and handles it at the office.

- Runs a first prior-art search
- Writes description, drawings, abstract and, above all, the **claims**: broad enough to protect, narrow enough to be valid

3 · Filing

The application is submitted to a patent office and the fees are paid (remember: to submit is not free).

- The office gives it an application number and a filing date
- The first filing fixes the date: the clock starts here

The first decision: where to file?

The problem: a patent is a **territorial** right. It protects only in the countries where it is granted, so the applicant must decide where protection is worth paying for.

WHAT DRIVES THE CHOICE? ?

Markets

Where will I sell, produce, or face competitors?

Budget

Fees, translations and attorneys, multiplied by every country

Time

How long can I wait before committing to countries?

Language and rules

Each office works in its own language and law

Rights are national, disclosure is global

Rights are national

One patent per country. To be protected abroad you must apply in each country, in its language.

Disclosure is global

A publication in any country enters the worldwide prior art: it "kills" novelty everywhere.

A special case: the EPO

One application at the EPO (in English, French or German), one examination, one grant decision. Then the patent is validated country by country.

Which patent office?

Patent offices grant patents; WIPO coordinates the international treaties.

WIPO

World Intellectual Property Organization, a UN agency in Geneva.

Administers the Paris Convention and the PCT. It does not grant patents.

EPO IP5

European Patent Office
Munich · Europe

USPTO IP5

US Patent and Trademark Office
Alexandria · USA

JPO IP5

Japan Patent Office
Tokyo · Japan

CNIPA IP5

China National IP Administration
Beijing · China

KIPO IP5

Korean IP Office
Daejeon · Korea

UIBM

Ufficio Italiano Brevetti e Marchi
Rome · Italy

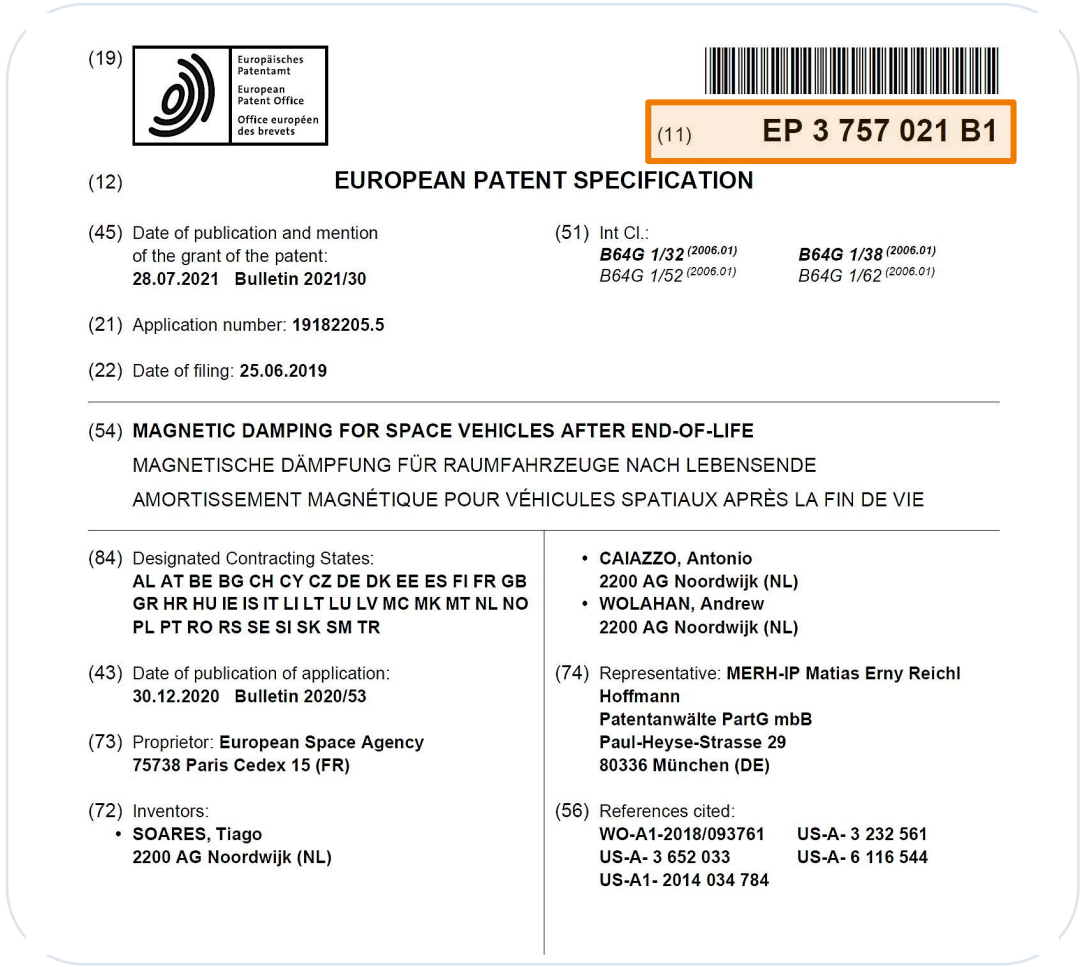
03

What does a patent document look like?

An example from the EPO

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The publication number and kind code



(19) Europäisches Patentamt
European Patent Office
Office européen des brevets

(11) EP 3 757 021 B1

(12) EUROPEAN PATENT SPECIFICATION

(45) Date of publication and mention of the grant of the patent:
28.07.2021 Bulletin 2021/30

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B64G 1/32 *(2006.01)* ***B64G 1/38*** *(2006.01)*
B64G 1/52 *(2006.01)* *B64G 1/62* *(2006.01)*

(21) Application number: **19182205.5**

(22) Date of filing: **25.06.2019**

(54) **MAGNETIC DAMPING FOR SPACE VEHICLES AFTER END-OF-LIFE**
MAGNETISCHE DÄMPFUNG FÜR RAUMFAHRZEUGE NACH LEBENSENDE
AMORTISSEMENT MAGNÉTIQUE POUR VÉHICULES SPATIAUX APRÈS LA FIN DE VIE

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(43) Date of publication of application:
30.12.2020 Bulletin 2020/53

(73) Proprietor: **European Space Agency**
75738 Paris Cedex 15 (FR)

(72) Inventors:
- **SOARES, Tiago**
  2200 AG Noordwijk (NL)
- **CAIAZZO, Antonio**
  2200 AG Noordwijk (NL)
- **WOLAHAN, Andrew**
  2200 AG Noordwijk (NL)

(74) Representative: **MERH-IP Matias Erny Reichl Hoffmann**
Patentanwälte PartG mbB
Paul-Heyse-Strasse 29
80336 München (DE)

(56) References cited:
WO-A1-2018/093761
US-A- 3 232 561
US-A- 3 652 033
US-A- 6 116 544
US-A1- 2014 034 784

(11) **Which patent document am I looking at?**
Publication number

(11) **EP 3 757 021 B1**

EP 3 757 021 B1 reads in three parts:

EP the office that published it (WO for PCT, US, CN, JP…)

3 757 021 the serial number of the publication

B1 the **kind code**: the type of document. At the EPO: **A1 application with search report, A2 without, A3 search report published later, B1 granted patent.**

One invention produces several documents: the A1 publication and the B1 grant are two different records. Link them through the application number, never count both.

Application number and filing date

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(21) (22) **When and where did it start?**

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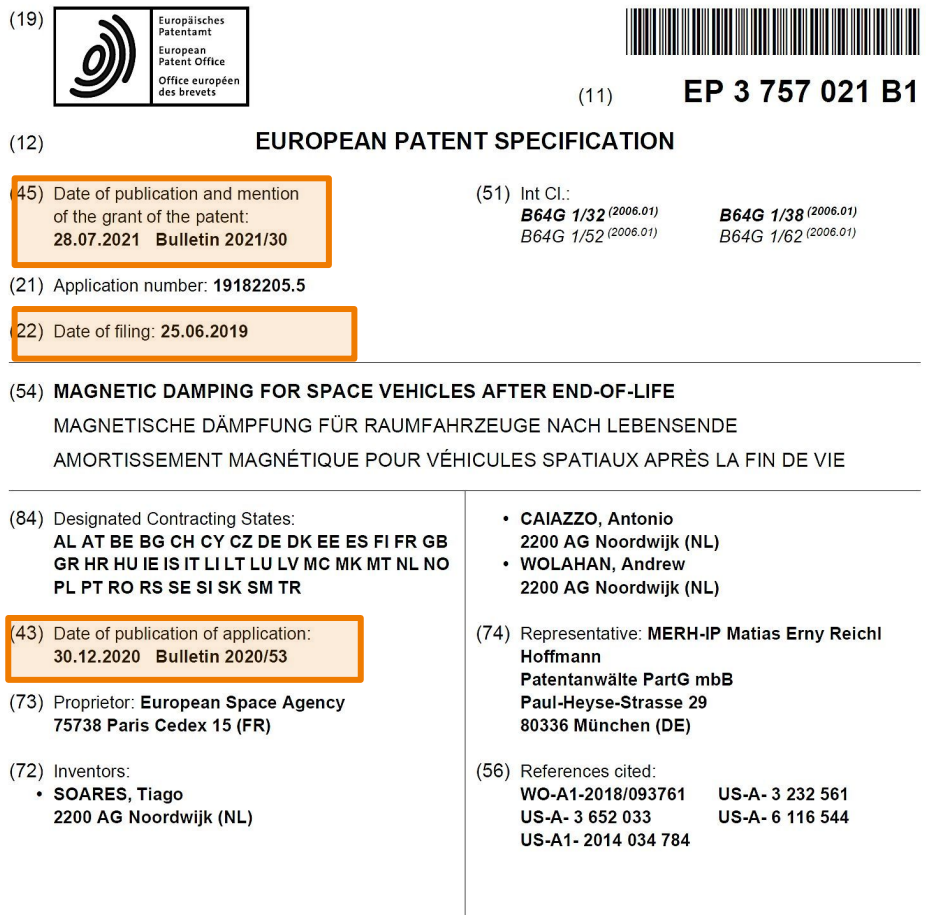
(22) Date of filing: **25.06.2019**

Application number 19182205.5: given by the EPO when the application arrives.

Date of filing 25.06.2019: the day the office received the application. The 20-year term of protection is counted from here.

The application number stays the same from filing to grant: it is the stable key that links all the publications of one application (in PATSTAT: appln_id).

How many dates related to a single patent?



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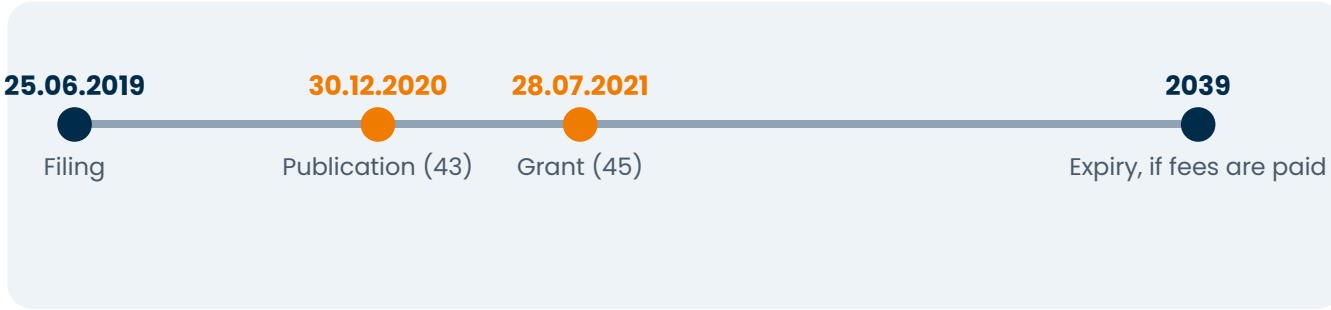
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(22) (43) (45) **When did each step happen?**

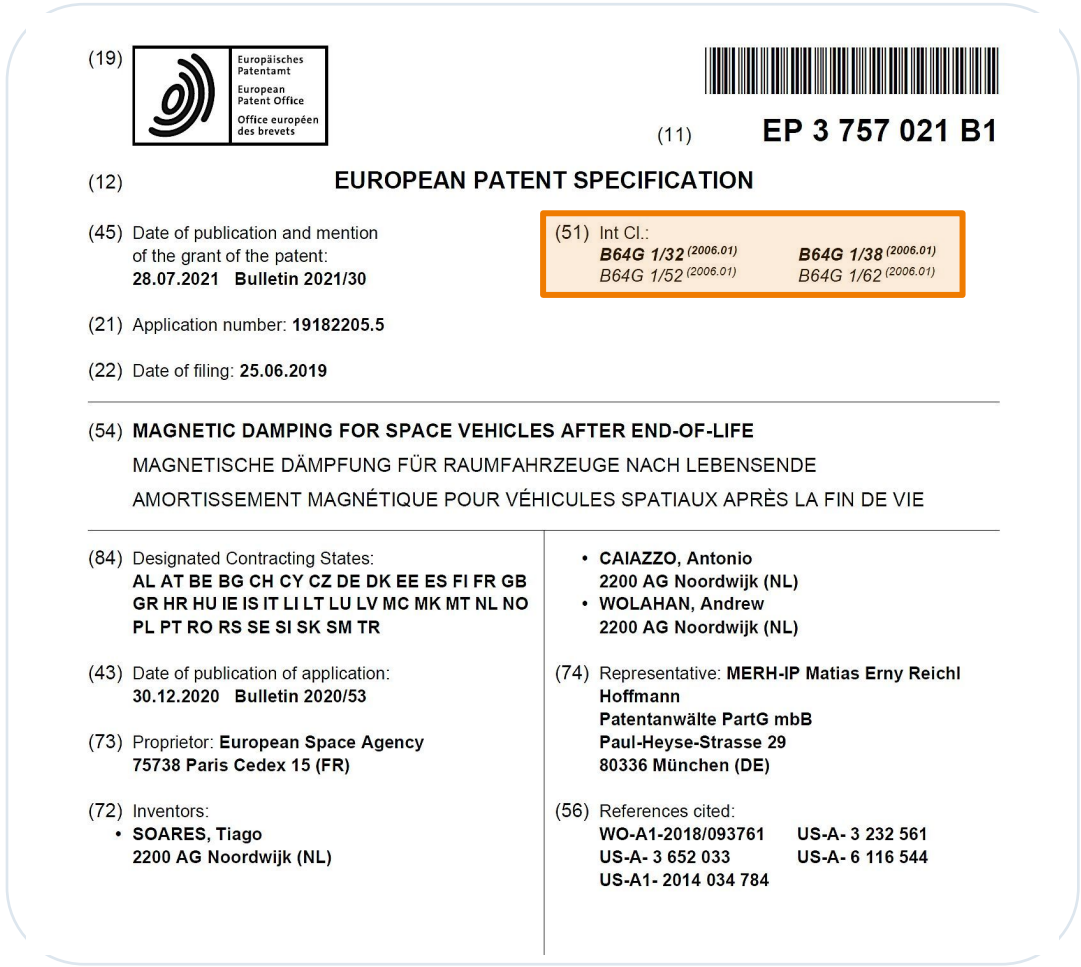


Filing date (22): when the application was received.

Publication (43) and grant (45): published 18 months after filing, granted about two years after filing.

Expiry: 20 years after filing date

The technological classification



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(51) Which technology is it?

(51) Int Cl.:
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B64G 1/52 *(2006.01)* *B64G 1/62* *(2006.01)*

Int. Cl. lists the International Patent Classification (IPC) codes given by the examiner: the technology fields the invention belongs to. The year in brackets is the version of the scheme.

Read B64G 1/32 from left to right: B section (performing operations; transporting) › B64 class (aircraft, aviation, cosmonautics) › B64G subclass (cosmonautics) › 1/32 group and subgroup, the specific technical feature.

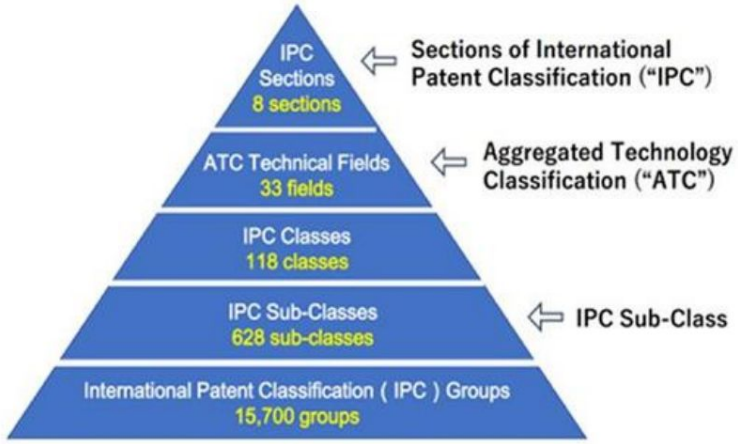
However, CPC classification is a much more granular version of IPC that is present in PATSTAT

Look codes up: https://www.cooperativepatentclassification.org/home

CPC vs IPC: patent classification

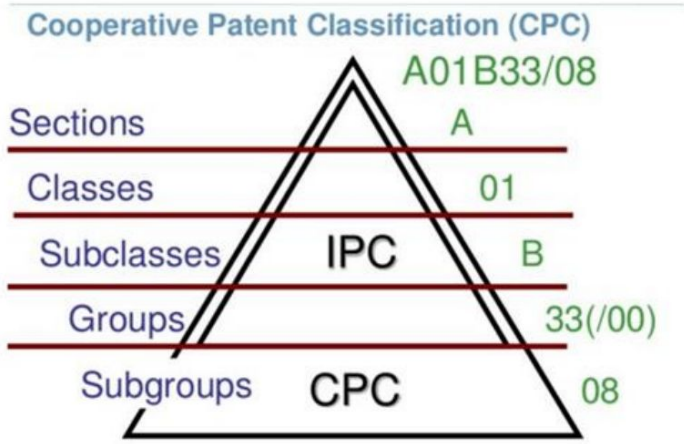
IPC · International Patent Classification

Managed by WIPO, used by over 100 offices.



CPC · Cooperative Patent Classification

An extension of the IPC with finer subdivisions, managed jointly by the EPO and the USPTO.



Reading a code

A	A01	A01B	A01B 33	A01B 33/08
section · Human necessities	class · Agriculture	subclass · Soil working	main group	subgroup

The title of the patent



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(54) What is it called?

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A short name for the invention, here in the three official languages of the EPO: English, German and French.

Titles are written by the applicant and are often deliberately generic: “Method and apparatus for…” says little about what is new.

Other parts of patent text reveal more details about the invention (explained in following lectures and tutorials about patent text)

EP only: the designated states



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(84) Where could it be protected?

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The **38 countries** of the **European Patent Convention** (in 2019) where the applicant may ask for protection.

Designation is not protection: after grant the owner chooses where to validate the patent, and pays for each country.

Who owns the patent?

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(73) **Who owns it?**

(73) Proprietor: **European Space Agency**
75738 Paris Cedex 15 (FR)

Applicant (71): who files the application and will own the patent. After grant the field becomes Proprietor (73): here the European Space Agency.

It can be a company, a university, a public body or an individual. Ownership can change later through sales and licences, recorded as legal events.

The “name game”: one owner appears under many spellings (Siemens AG, SIEMENS AKTIENGESELLSCHAFT, Siemens Healthcare…). Harmonise names before counting who leads a field, or use han_name in PATSTAT

The inventors



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(72) Who invented it?

- **CAIAZZO, Antonio**
2200 AG Noordwijk (NL)
- **WOLAHAN, Andrew**
2200 AG Noordwijk (NL)

The people who made the invention: Tiago Soares, Antonio Caiazzo and Andrew Wolahan, all at Noordwijk (NL), home of ESA's technical centre.

Inventors are usually employees and are not the owners. The right to be named belongs to them and cannot be transferred.

Inventor addresses show where the research is actually done, often not where the owner is registered. Co-inventor links form collaboration networks.

The representative

(19) Europäisches Patentamt / European Patent Office / Office européen des brevets

(11) **EP 3 757 021 B1**

(12) **EUROPEAN PATENT SPECIFICATION**

(45) Date of publication and mention of the grant of the patent:
28.07.2021 Bulletin 2021/30

(51) Int Cl.:
B64G 1/32 *(2006.01)* ***B64G 1/38*** *(2006.01)*
B64G 1/52 *(2006.01)* *B64G 1/62* *(2006.01)*

(21) Application number: **19182205.5**

(22) Date of filing: **25.06.2019**

(54) **MAGNETIC DAMPING FOR SPACE VEHICLES AFTER END-OF-LIFE**
MAGNETISCHE DÄMPFUNG FÜR RAUMFAHRZEUGE NACH LEBENSENDE
AMORTISSEMENT MAGNÉTIQUE POUR VÉHICULES SPATIAUX APRÈS LA FIN DE VIE

(84) Designated Contracting States:
AL AT BE BG CH CY CZ DE DK EE ES FI FR GB GR HR HU IE IS IT LI LT LU LV MC MK MT NL NO PL PT RO RS SE SI SK SM TR

(43) Date of publication of application:
30.12.2020 Bulletin 2020/53

(73) Proprietor: **European Space Agency**
75738 Paris Cedex 15 (FR)

(72) Inventors:
- **SOARES, Tiago**
  2200 AG Noordwijk (NL)
- **CAIAZZO, Antonio**
  2200 AG Noordwijk (NL)
- **WOLAHAN, Andrew**
  2200 AG Noordwijk (NL)

(74) Representative: **MERH-IP Matias Erny Reichl Hoffmann**
Patentanwälte PartG mbB
Paul-Heyse-Strasse 29
80336 München (DE)

(56) References cited:
WO-A1-2018/093761
US-A- 3 232 561
US-A- 3 652 033
US-A- 6 116 544
US-A1- 2014 034 784

(74) Who drafted it?

(74) Representative: **MERH-IP Matias Erny Reichl Hoffmann**
Patentanwälte PartG mbB
Paul-Heyse-Strasse 29
80336 München (DE)

The patent attorney firm that drafted the application and handled it at the EPO, here a firm in Munich.

This closes the loop of our story: the inventor's idea, written up by the attorney, filed at the office.

Representatives are rarely used in analyses, but they help identify who manages a firm's IP, and spot portfolios handled together.

The references cited



(19) Europäisches Patentamt
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US-A- 3 652 033 **US-A- 6 116 544**
US-A1- 2014 034 784

(56) What does it build on?

(56) References cited:
WO-A1-2018/093761 **US-A- 3 232 561**
US-A- 3 652 033 **US-A- 6 116 544**
US-A1- 2014 034 784

Earlier documents the examiner (or the applicant) found relevant: here one PCT application (WO-A1) and four US patents.

They come from the search report, coming up shortly. Each citation is a link from this patent to the prior art it builds on.

Backward citations (to the past) describe what an invention draws on; **forward citations** (received later) are a classic indicator of a patent's influence.

The text: abstract, description and claims

Behind the front page is the full text. Three parts, three different jobs.

1 Abstract

About 150 words summarising the invention.

- A search aid for examiners and analysts
- The most used text field in NLP studies (nb: NOT legal, abstract is not good for legal tasks, for legal tasks we shall use the claims)

2 Description

Explains the invention so that an expert could reproduce it. It may contain:

- Prior art, and its drawbacks
- The problem to be solved
- The solution and its advantages
- Ways of carrying it out, with drawings

3 Claims

Numbered sentences that define the legal scope of protection.

- **Independent claim(s):** the broadest definition
- Dependent claims: narrower versions that refer back

Try it: open patents.google.com/patent/EP3757021B1 and find the claims of the patent we just dissected.

18 months after filing: the **search report** is published

EP 2 256 022 A1

Europäisches Patentamt
European Patent Office
Office européen des brevets

EUROPEAN SEARCH REPORT

Application Number
EP 10 16 1440

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
A	EP 1 481 882 A (HONDA MOTOR CO LTD [JP]; KAYABA INDUSTRY CO LTD [JP]) 1 December 2004 (2004-12-01) * abstract; figure 14 *	1-16	INV. B62K21/08
A	DE 10 2006 024327 A1 (BAYERISCHE MOTOREN WERKE AG [DE]) 29 November 2007 (2007-11-29) * paragraph [0007] - paragraph [0008] *	1-16	
A	EP 1 318 069 A (HONDA MOTOR CO LTD [JP]; KABAYA INDUSTRY CO LTD [JP] HONDA MOTOR CO LT) 11 June 2003 (2003-06-11) * abstract; figure 2 *	1-16	

backward citations

The examiner lists the prior-art documents relevant to the application, with the most relevant passages and a letter code.

X Particularly relevant on its own: novelty or inventive step is at stake

Y Particularly relevant when combined with another document

A Technological background, not threatening

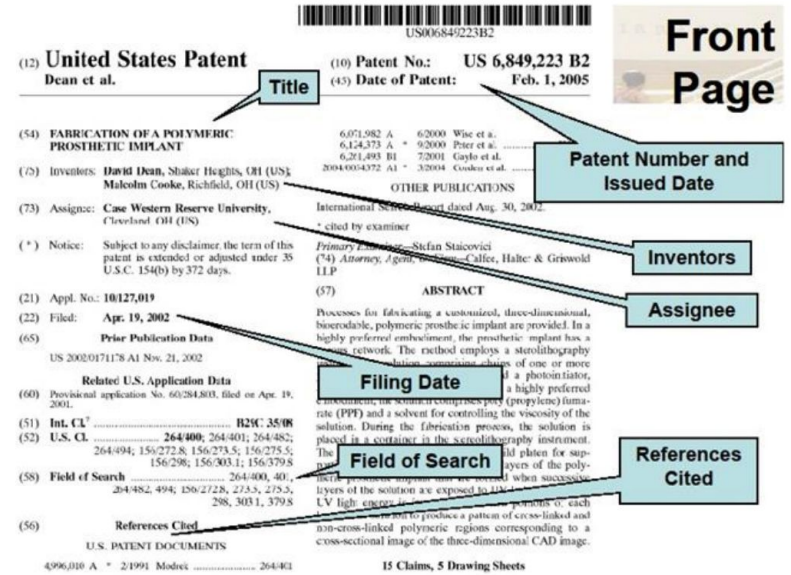
These references become the backward citations in patent databases.

The goal of filing: being granted

EP 2 256 022 B1 · the granted patent



A US patent front page (very similar to EP)



KIND CODES (EPO)

A1 application, with search report

A2 application, without search report

A3 search report published later

B1 granted patent

The same invention appears in the data more than once: A1 and B1 are two records.

Key patent features: a glossary

Inventor	The person who had the inventive idea. The status cannot be transferred, and omitting an inventor can invalidate the patent.
Applicant / assignee	Who files and owns the patent. Applicant= filing; assignee= owning the rights at a certain point of time
Abstract	About 150 words: a search aid for examiners and analysts.
Description	The prior art, the problem the invention solves, and at least one way to carry it out.
Claims	The legal heart of the patent: they define the extent of protection.
technological Classification	IPC / CPC codes: the technology field .
Application vs patent	A filed but not yet granted application is “pending”; a granted patent is final.
Search report	Issued by the office (EPO: about 6 months after filing): patentability assessment and blocking prior art.

The general path: from application to grant



Licensing: the owner can license from the day of filing. The right ends with refusal, revocation, unpaid renewal fees, or expiry.

Should applications be published before grant?

YES, because…

1. Inventors signal their inventions early, attracting licensees and potential investors
2. Rivals avoid duplicating R&D that is already under way
3. Knowledge spreads faster, fuelling cumulative innovation

NO, because…

1. Small inventors may be copied or “invented around” by larger rivals
2. Notice **pollution**: abandoned, low-quality applications still block others as prior art

04

Patent families

One invention, many countries, many documents

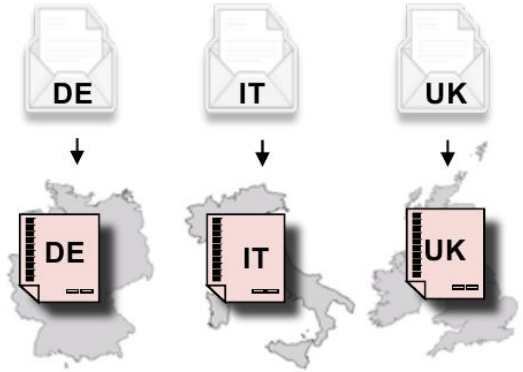
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Three ways to get protection

Rights are always national in the end. What changes is how many offices you deal with, and when.

1

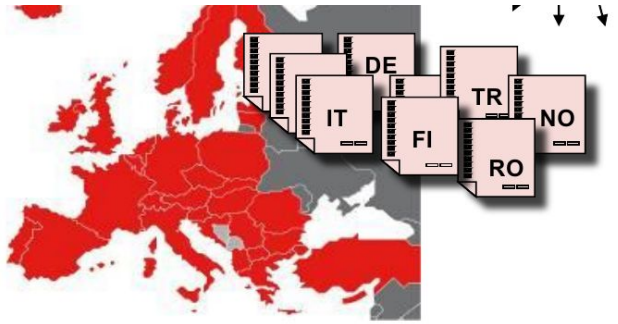
National route



One office per country, e.g. UIBM (Italy), DPMA (Germany), USPTO

2

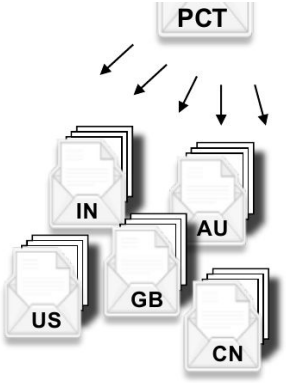
European route (EPC)



One application at the EPO, one examination, then a bundle of national patents

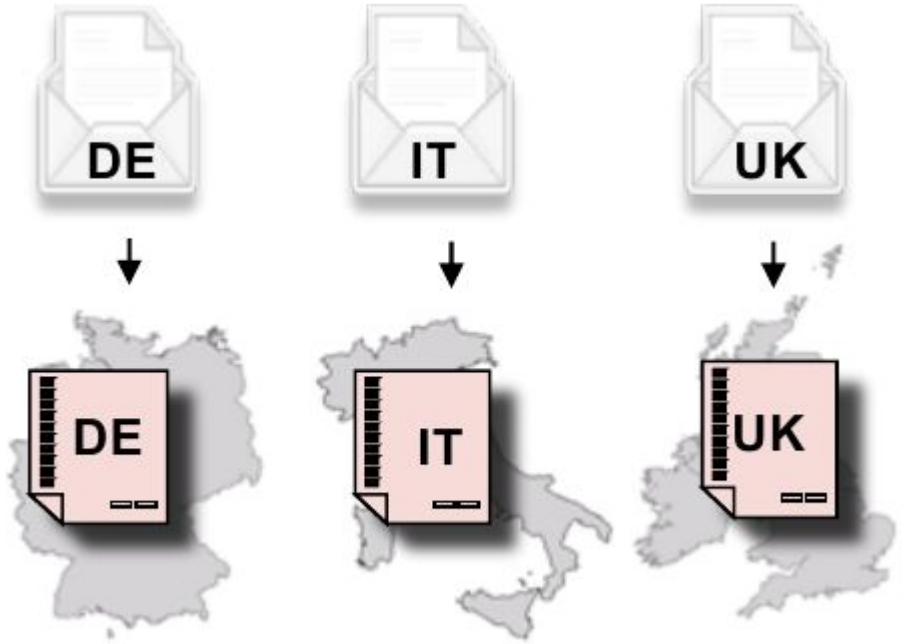
3

International route (PCT)



One application through WIPO for over 150 countries; decide where to go at month 30

The national route



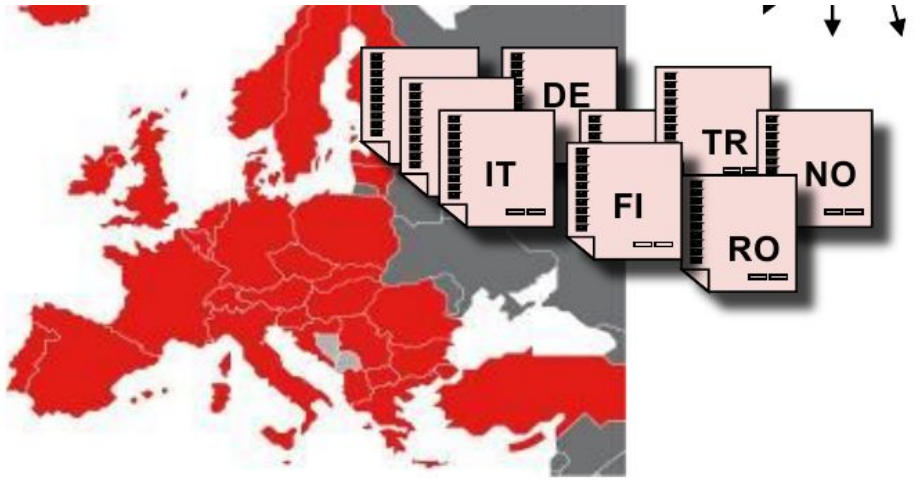
One application per country

- **Separate procedures:** each office searches, examines and grants on its own
- **First filing is the priority filing**
- **National law and language:** requirements, fees and translations differ
- **Costs add up** with every country, so it suits applicants who need only a few markets

After the first national filing you have 12 months to file in other countries while keeping the original priority date. Anything published in between does not destroy your novelty.

This is the "Paris route": national filings, linked by priority.

The European route (EPC)



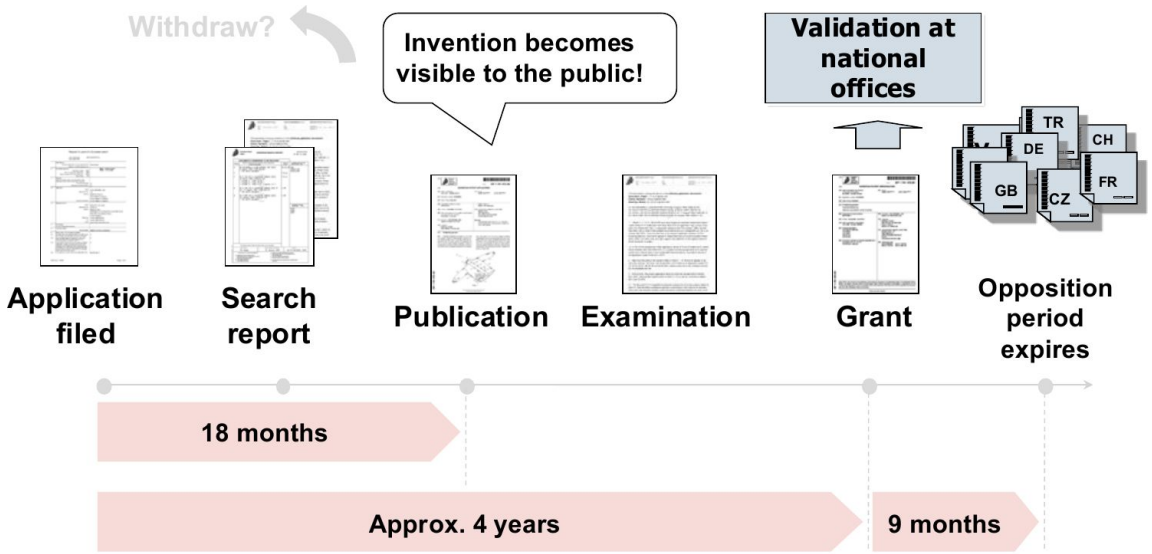
One application, one procedure

- **One application** at the EPO, in English, French or German
- **One search and one examination** for up to 39 member states, plus extension and validation states
- **The applicant chooses** the states where protection is wanted (designated states)
- **Result:** one European patent that becomes a bundle of national patents

Created by the European Patent Convention (1973). The EPO opened in 1977 in Munich.

The EPO validation

The grant procedure before the EPO



1 One procedure
Filing, search, publication at 18 months, examination: about 4 years in total.

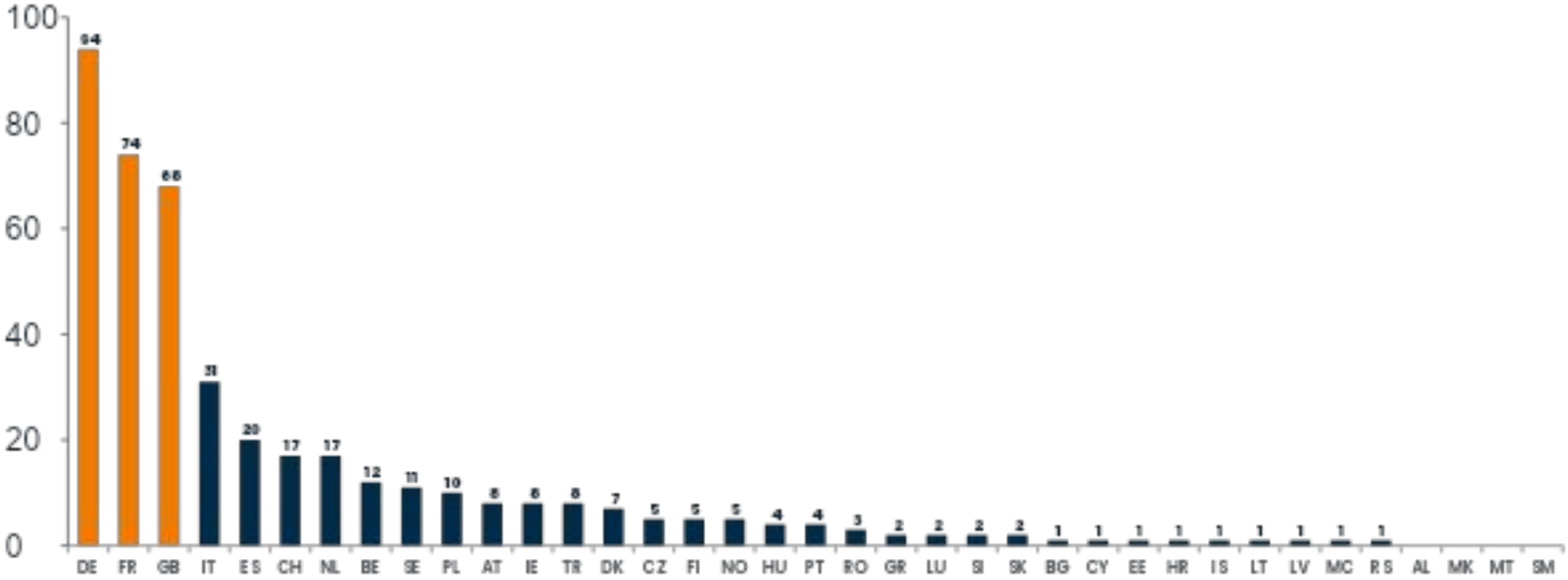
2 Grant
Granted once, by the EPO, and published in the European Patent Bulletin.

3 Validation
Within a short deadline (usually 3 months) the owner picks the states, and in each pays a fee and, where required, files a translation.

4 A bundle of national patents
National renewal fees every year, national courts, country-by-country lapses.

Where European patents are actually validated

Share of EPO-granted patents validated in each country (%)

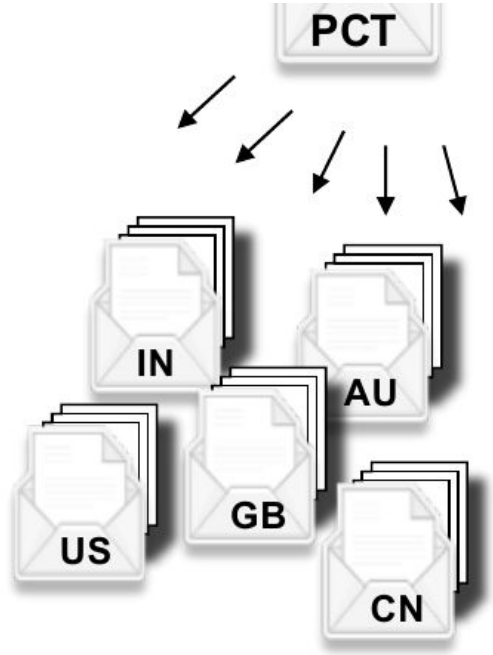


Source: own elaboration on PATSTAT (May 2024); patents granted June 2018 – May 2023.

The international route: PCT and WIPO

WIPO (World Intellectual Property Organization, a UN agency in Geneva) administers the Patent Cooperation Treaty (PCT, 1970). It does not grant patents: it coordinates one international application valid for over 150 countries.

Filing	Search	Publication	Examination	National phase
by 12 m	**~16 m**	**18 m**	**optional**	**30 m**
one international application at a receiving office, claiming the priority	international search report and written opinion on patentability	the application becomes public worldwide	a preliminary opinion before deciding	choose countries; each office decides on the grant



From the first filing to a family: two routes

Everything starts from one first filing. Within 12 months the applicant chooses how to go **abroad**; the deadlines depend on the route.

month 0 · month 12 · month 16 · month 18 · month 30 / 31

MONTH 0
First filing
e.g. at UIBM, Italy
sets the priority date

A Paris route

separate filings at each office, each claiming the priority

BY MONTH 12
File at each office
EPO, USPTO, CNIPA, JPO...: one application per office, each claiming the priority

AFTER MONTH 12
Each office works on its own
Each one searches, publishes (18 months after the priority) and examines. All countries must be chosen and paid for at month 12.

B PCT route

one international application first, countries chosen later

BY MONTH 12
One PCT application
filed once (WO), claiming the priority

~ MONTH 16
Search report
opinion on patentability

MONTH 18
Publication
public worldwide

MONTH 30 / 31
National / regional phase
choose the countries and pay: EPO 31 months, most offices 30

Paris route is faster, you get the patent before and is less expensive. However, you have less time to select the countries for protection compared to PCT route. PCT is more expensive, but more time to select countries and also is a single application, Paris is single applications in fact.

What is a patent family?

DEFINITION

A patent family is the set of patent applications, filed at different offices, that protect the same invention and claim the same priority (the first filing).

One invention = one family, however many documents and countries.



Why it matters: in patent databases each of these documents is a separate record. The family links them back to the one invention.

Vocabulary priority date: date of the first filing · extension: a filing abroad that claims that priority · family: all the filings that share it

TWO FAMILY DEFINITIONS IN DATABASES

Simple family (DOCDB): exactly the same priorities. (best to be used in the PW)

Extended family (INPADOC): linked by at least one shared priority, directly or through other documents. Larger.

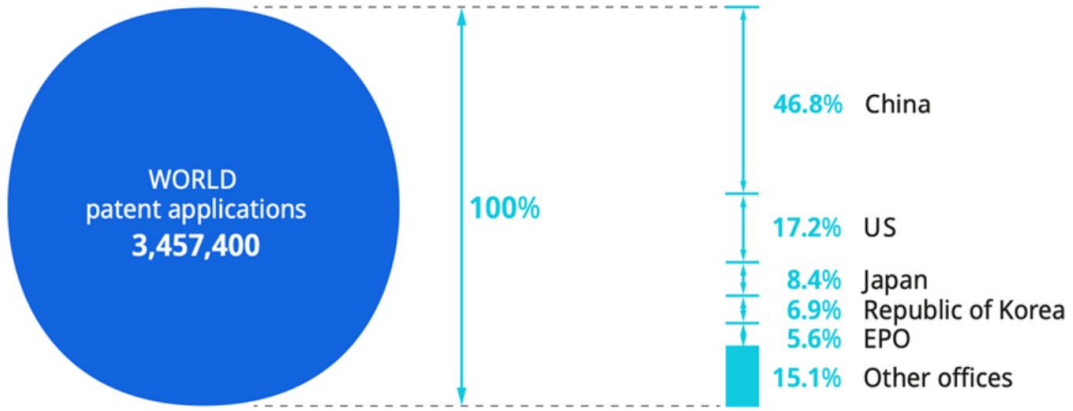
05

Some figures from the EPO

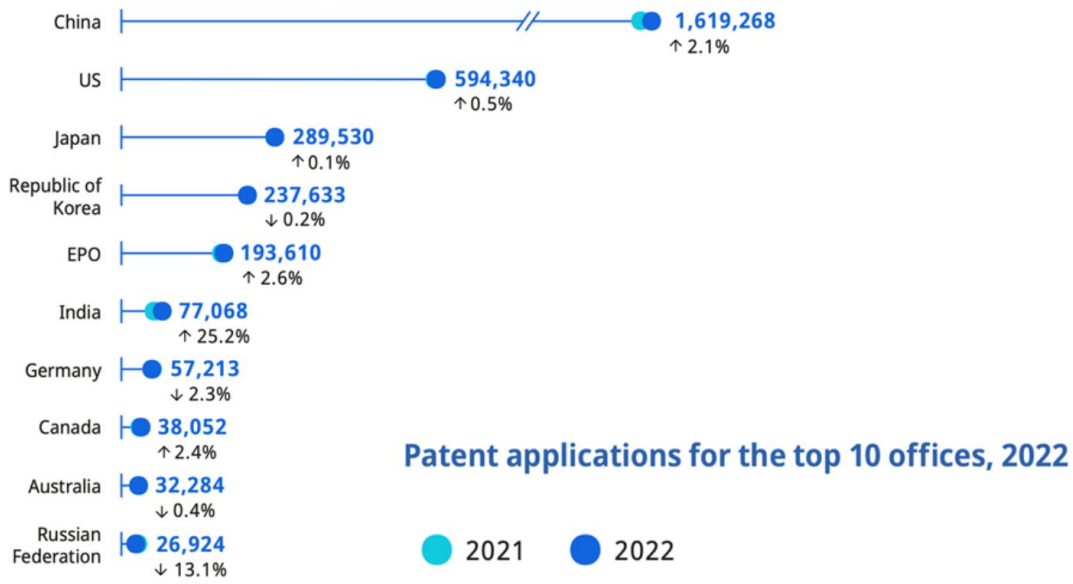
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The EPO among the world's patent offices

World patent applications, 2023: 3.46 million



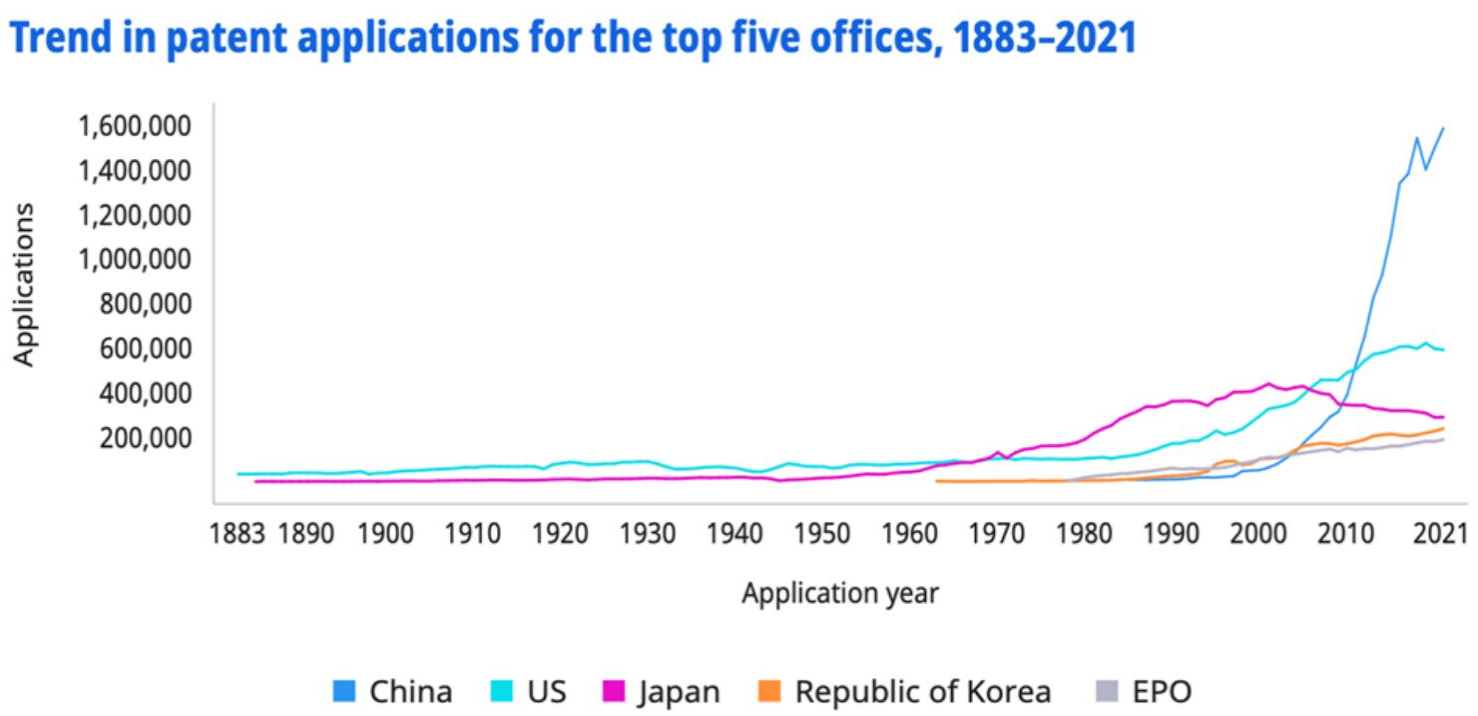
Patent applications at the top 10 offices, 2022



Take-away: the EPO is the fifth office in the world by applications (about 194,000 in 2022), after China, the US, Japan and Korea. Source: WIPO Statistics Database.

A century and a half of patent applications

Applications at the top five offices, 1883–2021



WHAT YOU SEE

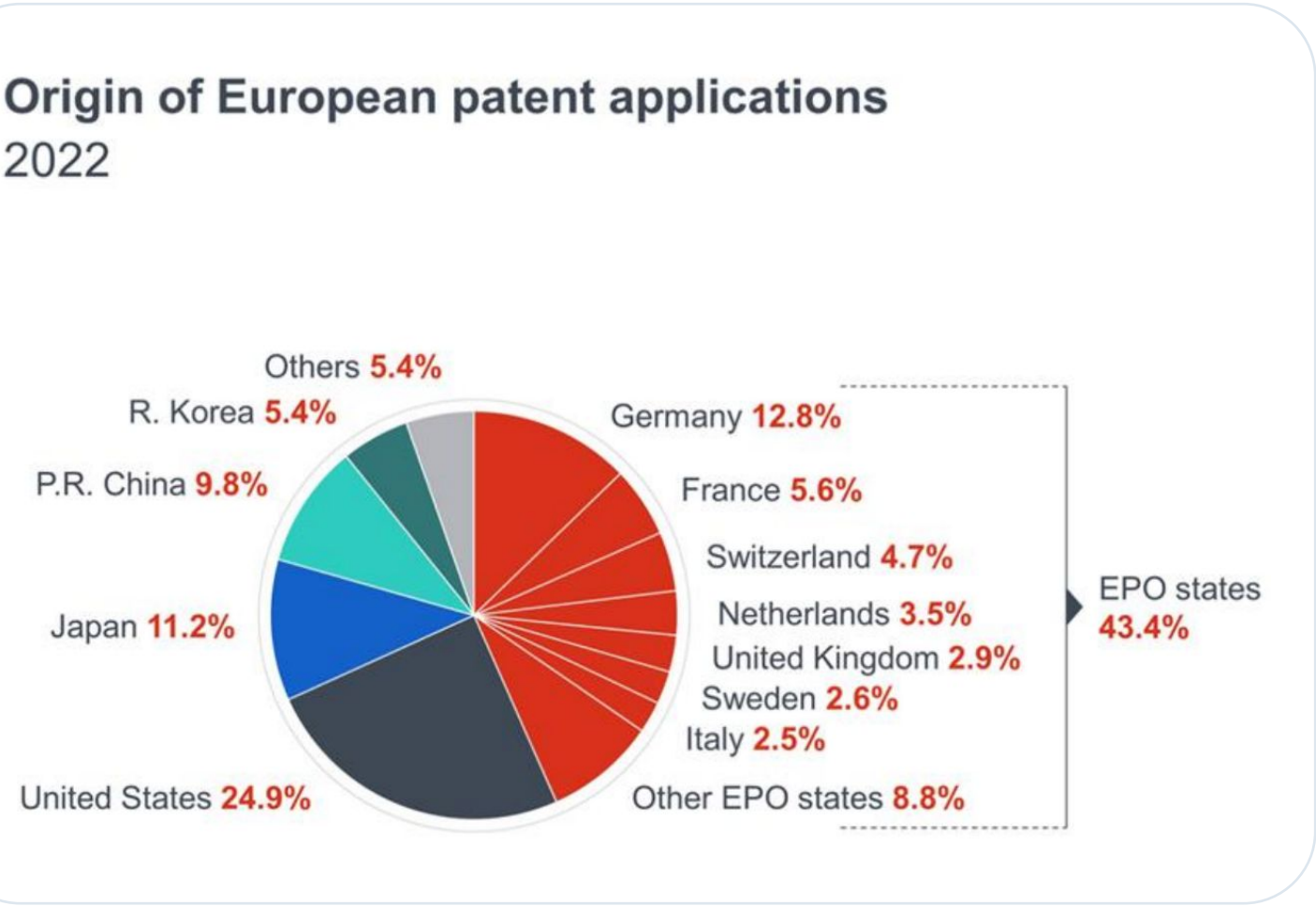
Japan
leads from the 1970s to the 2000s, then declines

United States
grows steadily since the 1980s

China
explodes after 2000 and becomes the largest office in the world

EPO
opens in 1978 and grows slowly: a regional office, much smaller than national giants

Who files at the EPO? Origin of applications, 2022



43.4% **from the 39 EPO member states**

Germany 12.8% · France 5.6% · Switzerland 4.7% · Netherlands 3.5% · UK 2.9% · Sweden 2.6% · Italy 2.5%

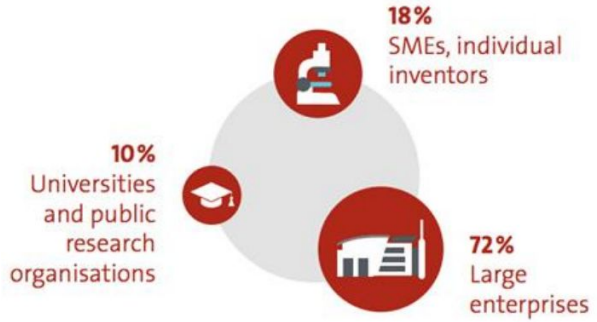
24.9% **from the United States**

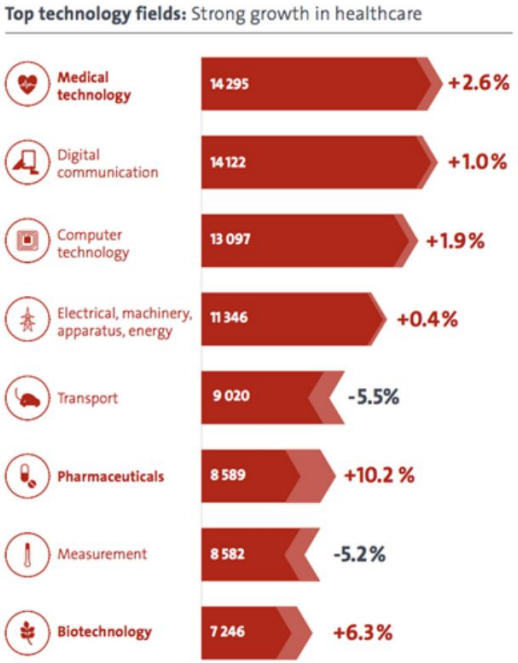
26.4% **from Asia**

Japan 11.2% · China 9.8% · Korea 5.4%

More than half of European applications come from outside Europe: the EPO is the gateway to the European market.

Who files, and in which technologies?





72% of applications come from large enterprises; 18% from SMEs and individual inventors; 10% from universities and public research

Medical technology is the top field (14,295 applications), just ahead of digital communication and computer technology

+10.2% growth in pharmaceuticals, the fastest among the top fields; transport falls by 5.5%

The top applicants at the EPO, 2022



- **ICT dominates the top:** Huawei (4,505 applications), LG, Qualcomm, Samsung and Ericsson lead the ranking
- **Non-European firms lead:** four of the top five come from China, Korea and the US; the first European applicant is Ericsson, fifth
- **Intangible assets** are the key driver of global competition

Source: EPO Patent Index 2022.

The Unitary Patent (since 1 June 2023): what changes

After grant, the owner of a European patent can now request “unitary effect” within one month, at no fee.

	European patent	Unitary patent
Territory	The states you validate, one by one	**All participating EU states at once (18 today; Spain and Croatia are out)**
Translations	Required by many states at validation	**One translation only**
Renewal fees	Paid to each national office	**One fee to the EPO: cheaper if you need 4 or more countries**
Courts	National courts, possibly with different outcomes	**The Unified Patent Court: one decision for all states**
Revocation	Country by country	**Everywhere at once**
Flexibility	You can drop countries later to save fees	**All or nothing: no dropping countries later**

06

Challenging a patent (after grant)

Opposition at the EPO, post-grant review at the USPTO

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Opposition at the EPO: how it works

What it is: an administrative procedure at the EPO to revoke or limit a granted European patent, centrally, for all states at once, without going to court.

Who

Anyone except the owner, even a "straw man" filing on behalf of someone who wants to stay anonymous.

When

Within **9 months of the publication of the grant**.

On what grounds

The invention is **not patentable** (no novelty, no inventive step…), it is not **disclosed clearly enough to be reproduced**, or the patent goes beyond the application as filed.

Notice of opposition and fee → **Opposition division: 3 EPO examiners** → **Written exchange of arguments** → **Oral proceedings** → **Decision** → **Appeal to the Boards of Appeal**

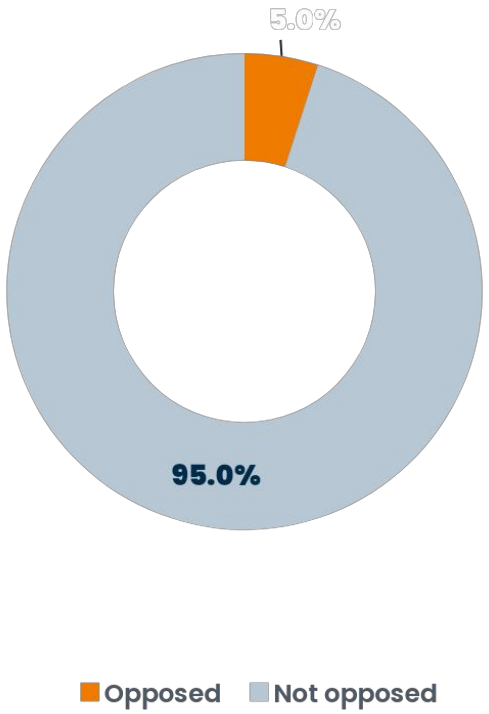
Three possible outcomes: the opposition is **rejected**, the patent is **revoked**, or it is **maintained in amended (narrower) form**. Valid in all states.

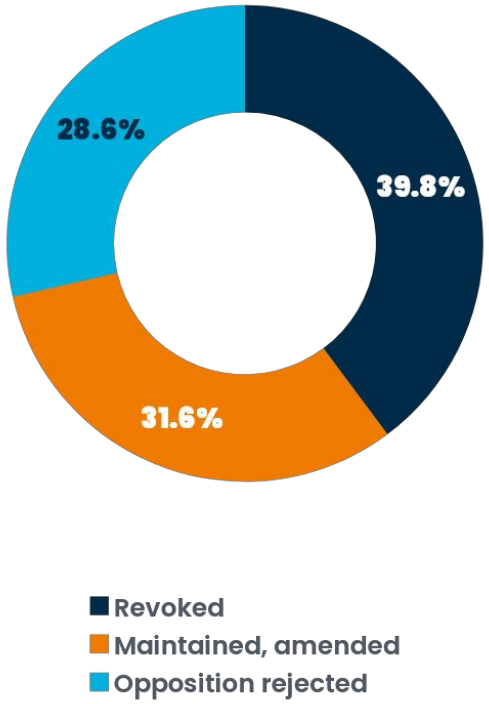
Challenging a patent at the EPO: key facts and numbers

- **Who:** anyone except the owner, even a “straw man” acting for someone else
- **When:** within 9 months of grant
- **Grounds:** lack of novelty, lack of inventive step, insufficient disclosure
- **Outcomes:** rejected, patent revoked, or maintained in amended form
- **Duration:** about 34 months on average
- **Cost:** €880 fee; €5,000–10,000 in total for simple cases

WHEN DO YOU THINK PATENTS ARE OPPOSED?

EPO oppositions, 2018





Challenging a patent at the USPTO: the PTAB

Since 2012, the Patent Trial and Appeal Board (PTAB) runs two kinds of trials. Compared with **EPO opposition**:

	EPO OPPOSITION	POST GRANT REVIEW	INTER PARTES REVIEW
When	Within 9 months of grant	Within 9 months of grant	After the 9-month PGR window
Who	Anyone, even anonymously via a straw man	Must name all real parties in interest	Must name all real parties in interest
Grounds	Novelty, inventive step, insufficient disclosure	Novelty, obviousness, written description or enablement	Only prior art (patents, printed publications): novelty and obviousness
Cost	€880 fee; €5–10k in total	~$20k fee; $100–250k in total	$50–150k in total

Much rarer in the US: about 5% of EPO patents are opposed, vs 0.46% of US patents challenged at the PTAB (10,999 of 2,367,153 granted in 2013–2019).

07

Anticipation of next lecture



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Other legal disputes related to patents

	INVALIDITY	INFRINGEMENT
Question	Should this patent exist at all?	Is someone using what the claims protect without permission?
Who starts it	Anyone who feels threatened by the patent	The patent owner, against the alleged infringer
Where	Patent office (opposition, PGR, IPR) or court	Court only
When	At the office: time limits. In court: any time	Whenever infringement happens, during the patent's life
Typical cost	Court: often above $500,000 in the US	From tens of thousands to millions

In practice the two travel together: sued for infringement, the defendant almost always counter-attacks the validity of the patent.

Questions?

Tomorrow
PW data collection: the TIP platform · Tuesday 13 October, 17:30–19:00 · Room R3

Next lecture
IPR as a strategic choice for companies · Monday 19 October, 13:00–14:30 · Room 4P

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